Maximal divergence of the leximin and leximax core selections in exact games

Maxwell Flanagan

Draft of August 7, 2026

Abstract

A singleton-normalized exact transferable-utility game carries two canonical egalitarian core selections: the leximin allocation Lmin and the leximax-min allocation Lmax. They coincide on every exact game with at most four players. We measure their separation by $\rho(v) = \|\operatorname{Lmin}(v) - \operatorname{Lmax}(v)\|_1/v(N)$, which never exceeds 2, and show that 2 is the exact supremum. The witnesses form an explicit two-parameter family $G(p,k)$ of two-generator lower-envelope games with $\rho(G(p,k)) = 2(p(k-1)-1)/(k(p+1))$ and the deficit identity $2-\rho(G(p,k)) = 2n/(k(p+1))$, $n = p+1+k$, so that $s(n) \geq 2 - 8n/(n^2 - (n \bmod 2))$ for every $n \geq 4$; the ratio reaches 1 at eight players and exceeds it from nine on. The mechanism is a single family of tight $(p+1)$-player caps that both selections saturate: leximin reads a cap as a ceiling and piles the residual onto one player, leximax-min reads it as a budget and spreads the residual over half the players, so the two supports become asymptotically disjoint. The concentrating selection is the leximin one. On the two-generator class we prove a dual description of the core, $\operatorname{Core}(v) = \{x \geq 0 : x(N) = T,\ x(S) \leq \max(g(S), h(S))\}$, deduce a one-coalition upper bound on the leximin floor, and exhibit the smallest game in a complete enumeration at which the mirror bound for the leximax-min ceiling fails. At $n = 5$ the family formula is not sharp: a five-player game analysed by hand attains $1/2$, the two-generator enumeration — complete for small denominators — finds nothing above it, and we prove that at five players *both* egalitarian levels are certified by single coalitions — the property whose failure from six players on accompanies the climb of $s(n)$. We also close one natural route to a matching upper bound, by a seven-player game in which smoothing a generator toward uniformity collapses ρ from $1/5$ to 0.

1 Introduction

Let (N, v) be a transferable-utility game on $N = \{1, \ldots, n\}$ with

$$\operatorname{Core}(v) = \{x \in \mathbb{R}^N : x(N) = v(N),\ x(S) \geq v(S) \text{ for all } S \subseteq N\},$$

where $x(S) = \sum_{i \in S} x_i$. The game is *exact* if every coalition inequality is attained by some core allocation, and *singleton-normalized* if $v(\{i\}) = 0$ for every i; on that domain every core allocation is nonnegative. Exact games are the natural home for egalitarian core selections, and two such selections are canonical. The *leximin* allocation $\operatorname{Lmin}(v)$ maximizes the increasing rearrangement of x lexicographically over the core; the *leximax-min* allocation $\operatorname{Lmax}(v)$ minimizes the decreasing rearrangement lexicographically over the core. Both are single-valued on a nonempty compact convex set. The first raises the poorest as far as possible, the second lowers the richest as far as possible.

On small games the two agree. The companion paper (Flanagan, 2026) proves that every singleton-normalized exact game on at most four players is an exact partition game at Lmin; by

the established implication from exact partition to Lorenz domination (Llerena and Mauri, 2017), Lmin then Lorenz-dominates every core allocation, and a Lorenz-greatest core point is simultaneously leximin- and leximax-optimal[1], so $\mathrm{Lmin}(v) = \mathrm{Lmax}(v)$ for every singleton-normalized exact game with $n \leq 4$. Our complete enumerations at $n \leq 4$ reconfirm this rigidity as a control (Section 5). The adjacent large-core chain — exact games on at most four players have large cores (Biswas et al., 1999), and on large-core games the procedural egalitarian solution coincides with Lmax (Dietzenbacher, 2021; Arin et al., 2003) — pins only the Lmax side: it does not deliver the coincidence, and indeed a five-player exact game with a large core has $\mathrm{Lmin} \neq \mathrm{Lmax}$: the lower envelope of $(2, 0, 12, 6, 0)$, $(0, 0, 0, 0, 20)$ and $(0, 2, 10, 4, 4)$, with $\| \mathrm{Lmin} - \mathrm{Lmax} \|_1 = v(N)/5$ (verified exactly; archived with the repository). The companion's lower-envelope realization technique is used again here.

The pair itself has recent precedent. Vazirani (2024) and Gangam et al. (2024) study leximin and leximax core imputations of assignment, flow and matching games, observe on examples that the two selections differ, and describe their opposition in exactly the direction this paper quantifies; the leximax selection appears earlier as the leximax solution of neighbour games (Klijn et al., 2003) and as the lexmax solution in the egalitarian-set literature (Llerena and Mauri, 2016), besides its large-core characterization (Arin et al., 2003). What is new here is the quantitative worst case: the exact domain, the supremum 2 with its support-disjointness characterization, the closed-form extremal family, and the direction result that the *concentrating* selection is the leximin one.

The question of this paper is what happens after the rigidity breaks. Write

$$\rho(v) = \frac{\| \mathrm{Lmin}(v) - \mathrm{Lmax}(v) \|_1}{v(N)}, \qquad s(n) = \sup\{\rho(v) : \ v \text{ exact on } n \text{ players}, \ v(N) > 0\},$$

the supremum running over singleton-normalized games. Since $\mathrm{Lmin}(v)$ and $\mathrm{Lmax}(v)$ are nonnegative with the same total $v(N)$, we have $\rho \leq 2$ always, and $\rho = 2$ would mean disjoint supports. The rigidity says $s(n) = 0$ for $n \leq 4$. How large can $s(n)$ get?

Results. The answer is: as large as arithmetically possible.

- $\sup_n s(n) = 2$, and at fixed n, $s(n) \geq 2 - 8n/(n^2 - (n \bmod 2))$, which is $2 - 8/n$ for even n (Theorem 17); in particular $s(n) \geq 1$ from $n = 8$ and $s(n) > 1$ from $n = 9$, and the bound is attained on the two-generator class at $n = 6, 7, 8$ by complete enumeration.
- The witnesses form a two-parameter family $G(p, k)$ whose two allocations are computed in closed form (Theorem 13), with the deficit identity $2 - \rho(G(p, k)) = 2n/(k(p + 1))$ (Corollary 15).
- On the two-generator class the core has a dual description as a system of upper caps (Theorem 6), from which the leximin floor inherits a one-coalition certificate (Theorem 7); the mirror statement for the leximax-min ceiling is false, and we give the smallest counterexample (Observation 10).
- $n = 5$ is anomalous: the family formula gives $1/3$, a game analysed by hand attains $1/2$ (Theorem 21), the two-generator enumeration (complete for small denominators) finds nothing above it, and the $n = 5$ extremal class is qualitatively unlike the $n \geq 6$ one.

The mechanism, in one sentence. In $G(p, k)$ a single family of tight caps forces $p + 1$ players to share only p; leximin reads each cap as a ceiling, so the floor sticks at $p/(p + 1)$ and the entire

[1] With equal totals, Lorenz domination gives $\sum_{i \leq k} x_i^{\uparrow} \geq \sum_{i \leq k} y_i^{\uparrow}$ for every k; induction on k yields $x^{\uparrow} \geq_{\mathrm{lex}} y^{\uparrow}$, and complementing partial sums gives the leximax-min half. See Dutta and Ray (1989) for the Lorenz-dominant core element on convex games.

residual piles onto one player, while leximax-min reads the same cap as a budget, loading half the players up to $p(k-1)/k$ and thereby squeezing the capped players down to $1/k$ — so the two supports become asymptotically disjoint and $\rho \to 2$. It is worth stating the direction explicitly, because it inverts the usual reading: the *concentrating* selection is the leximin one. In $G(p,k)$ the leximin allocation gives one player a share $p(k-1)/((p+1)k) \to 1$ of $v(N)$ while everyone else sits just below 1; it is nevertheless leximin-optimal, because nothing better is feasible for the poorest. Both selections are driven into extreme shapes by the same constraint, in opposite directions.

What is not proved. An upper bound matching Theorem 17 at fixed n is open, and Section 6 shows that one natural route to it is closed. Section 7 separates the proved statements from the enumerated and the conjectured ones, and lists the open problems.

2 Preliminaries

Throughout, $N = \{1, \dots, n\}$, all games are singleton-normalized, and $T := v(N) > 0$. Vectors are indexed by N, and $\operatorname{supp}(x) = \{i : x_i \neq 0\}$.

2.1 Exact games as lower envelopes

The following generation fact is the only construction used in this paper. It is standard; see Csóka et al. (2011) and Studený and Kratochvíl (2022) for the surrounding theory of the exact cone, and Flanagan (2026) for the same statement in the companion.

Lemma 1 (lower-envelope generation)**.** *Let $g_1, \dots, g_r \in \mathbb{R}^N_+$ have a common total $g_1(N) = \cdots = g_r(N) = T$, and set $v(S) = \min_j g_j(S)$. Then v is exact, $v(N) = T$, every g_j lies in* Core(v)*, and v is singleton-normalized as soon as every coordinate vanishes in some g_j. Conversely, every exact singleton-normalized game is of this form: one may take $g_1, \dots, g_r$ to be the vertices of* Core(v)*.*

Proof. Each g_j satisfies $g_j(S) \geq v(S)$ by definition of the minimum and $g_j(N) = T = v(N)$, so $g_j \in \operatorname{Core}(v)$; and the inequality at S is attained by whichever g_j realizes the minimum, so v is exact. The singleton statement is immediate from $v(\{i\}) = \min_j (g_j)_i$. Conversely, if v is exact then $v(S) = \min_{x \in \operatorname{Core}(v)} x(S)$ for every S, and the minimum of a linear functional over the polytope Core(v) is attained at a vertex; the vertices are nonnegative because v is singleton-normalized, and all have total T. □

We call a game generated as in Lemma 1 by r allocations an *r-generator* game. Part of what follows is specific to $r = 2$.

2.2 The two selections

For $x \in \mathbb{R}^N_+$ let $x^{\uparrow}$ be its increasing and $x^{\downarrow}$ its decreasing rearrangement.

Definition 2. Lmin(v) *is the allocation in* Core(v) *whose increasing rearrangement is lexicographically largest;* Lmax(v) *is the allocation in* Core(v) *whose decreasing rearrangement is lexicographically smallest.*

Both exist and are unique on a nonempty compact convex set: existence is compactness applied stage by stage, and uniqueness follows by comparing two putative optima with their midpoint, which is feasible by convexity and strictly better at the first stage where they differ. We write $\mu(v) = \min_i \operatorname{Lmin}(v)_i$ for the *leximin floor* and $M(v) = \max_i \operatorname{Lmax}(v)_i$ for the *leximax-min ceiling*; equivalently $\mu = \max_{x \in \operatorname{Core}} \min_i x_i$ and $M = \min_{x \in \operatorname{Core}} \max_i x_i$.

2.3 Normalization and the ratio

Two conventions have to be stated honestly, because neither selection is translation-equivariant: both are applied to the raw payoff vector x, not to $x - (v(\{i\}))_{i\in N}$, and on a game that is not singleton-normalized these are different objects — the leximax-min allocation in particular changes under the shift. We therefore fix the domain once and for all: all games are singleton-normalized, and both selections are computed on the normalized game. No claim below asserts invariance of Lmin or Lmax under adding singleton values.

Scaling, by contrast, is harmless: $\mathrm{Core}(cv) = c\,\mathrm{Core}(v)$ for $c > 0$ and both selections are positively homogeneous, so $\mathrm{Lmin}(cv) = c\,\mathrm{Lmin}(v)$ and $\mathrm{Lmax}(cv) = c\,\mathrm{Lmax}(v)$. Hence $\rho(v) = \|\,\mathrm{Lmin}(v) - \mathrm{Lmax}(v)\|_1 / v(N)$ is scale-invariant, which is what makes it comparable across games of different size. We record the trivial ceiling and the form of the deficit.

Lemma 3 (ceiling and overlap)**.** *For any singleton-normalized exact v with $T = v(N) > 0$,*

$$\rho(v) = 2\Big(1 - \frac{1}{T}\sum_{i\in N} \min\big(\mathrm{Lmin}(v)_i, \mathrm{Lmax}(v)_i\big)\Big) \;\le\; 2,$$

with $\rho(v) = 2$ if and only if $\mathrm{Lmin}(v)$ *and* $\mathrm{Lmax}(v)$ *have disjoint supports.*

Proof. For nonnegative x, y with $x(N) = y(N) = T$ one has $|x_i - y_i| = x_i + y_i - 2\min(x_i, y_i)$, and summing gives $\|x - y\|_1 = 2T - 2\sum_i \min(x_i, y_i)$. The overlap term is nonnegative and vanishes exactly when no coordinate is positive in both. □

So $\rho \to 2$ is precisely asymptotic support disjointness of the two egalitarian selections, and the whole content of the paper is that this can be forced.

3 The two-generator class

Throughout this section v is the lower envelope of exactly two nonnegative allocations g, h with common total T, so that $v(S) = \min(g(S), h(S))$ for every S (including $S = N$, where both sides equal T).

Lemma 4 (support disjointness)**.** *If v is singleton-normalized then $A := \mathrm{supp}(g)$ and $B := \mathrm{supp}(h)$ are disjoint, and every player outside $A \cup B$ is null, receiving* 0 *in every core allocation. Deleting null players changes neither* Lmin, Lmax*, nor T, so we may and do assume $A \cup B = N$ and $A \cap B = \emptyset$.*

Proof. $v(\{i\}) = \min(g_i, h_i) = 0$ forces $g_i = 0$ or $h_i = 0$, which is exactly $A \cap B = \emptyset$. If $i \notin A \cup B$ then $g_i = h_i = 0$, so $v(N \setminus \{i\}) = \min\big(g(N \setminus \{i\}), h(N \setminus \{i\})\big) = T$; any $x \in \mathrm{Core}(v)$ has $x(N \setminus \{i\}) \ge T$ and $x(N) = T$ with $x \ge 0$, hence $x_i = 0$. Both selections assign a null player 0, so deletion is harmless. □

Definition 5. *For $S \subseteq N$ put $f(S) := \max\big(g(S), h(S)\big) = \max\big(g(S \cap A), h(S \cap B)\big)$.*

3.1 A dual description of the core

Theorem 6 (two-generator core duality)**.**

$$\mathrm{Core}(v) = \big\{x \ge 0 :\; x(N) = T \text{ and } x(S) \le f(S) \text{ for all } S \subseteq N\big\}.$$

Proof. For x with $x(N) = T$ and any S, the inequality $x(S) \geq v(S)$ is equivalent to $x(N \setminus S) \leq T - v(S)$. It therefore suffices to prove

$$T - v(S) = f(N \setminus S) \qquad \text{for every } S \subseteq N. \tag{1}$$

Since g and h are modular with $g(N) = h(N) = T$,

$$T - v(S) = T - \min\big(g(S), h(S)\big) = \max\big(T - g(S),\, T - h(S)\big) = \max\big(g(N \setminus S),\, h(N \setminus S)\big) = f(N \setminus S).$$

Replacing S by $N \setminus S$ in (1) turns the family $\{x(S) \geq v(S)\}$ into the family $\{x(S) \leq f(S)\}$. Finally $x \geq 0$ is exactly the family of singleton inequalities $x(\{i\}) \geq v(\{i\}) = 0$. □

Identity (1) uses only modularity and the common total; the disjointness of Lemma 4 is not needed for it, and enters only to normalize the presentation of f. Note also that $S = N$ gives $f(N) = T$, so the cap at N is implied by efficiency, and $S = \emptyset$ gives $f(\emptyset) = 0$.

In fact the duality is fully general: for *any* singleton-normalized game, setting $f(S) := T - v(N \setminus S)$, the same complementation involution gives $\mathrm{Core}(v) = \{x \geq 0 : x(N) = T,\ x(S) \leq f(S)\ \forall S\}$. The two-generator content of Theorem 6 is the computation $f = \max(g, h)$. The general form is used in Section 5.4.

3.2 The leximin floor

Theorem 7 (leximin floor: the easy direction)**.** *Let*

$$\lambda := \min_{\emptyset \neq S \subseteq N} \frac{f(S)}{|S|} = \min_{(A', B') \neq (\emptyset, \emptyset)} \frac{\max\big(g(A'), h(B')\big)}{|A'| + |B'|},$$

the second minimum running over $A' \subseteq A$, $B' \subseteq B$. Then $\mu \leq \lambda$.

Proof. $\mathrm{Lmin}(v)$ is a core allocation, so by Theorem 6, for every nonempty S, $|S|\,\mu \leq \mathrm{Lmin}(v)(S) \leq f(S)$, i.e. $\mu \leq f(S)/|S|$. Take the minimum over S. □

The bound is a *one-coalition certificate*: a single tightest cap already pins the floor from above. Its natural habitat is polymatroid theory: when f is monotone submodular, the polytope of Theorem 6 is a polymatroid base polytope, the leximin floor *equals* λ, and the decreasingly-minimal and increasingly-maximal points coincide — the theory of lexicographically optimal bases (Fujishige, 1980), where $\rho = 0$ identically. The two-generator $f = \max(g, h)$ is not submodular in general; the conjecture below asks whether the floor identity survives the loss of submodularity, and the rest of the paper measures exactly how far the dec-min/inc-max coincidence fails. Empirically the bound is exact.

Conjecture 8 (leximin floor formula)**.** *$\mu = \lambda$ for every singleton-normalized two-generator exact game.*

Evidence and reduction. Conjecture 8 holds in 586 of 586 random two-generator games at $n = 4, 5, 6, 7$ (sample sizes 120, 218, 165, 83), with all arithmetic exact. What remains to prove is a clean linear-programming statement. Equality means that the system

$$y \geq 0, \qquad y(N) = T - n\lambda, \qquad y(S) \leq f'(S) := f(S) - \lambda|S|$$

is feasible; indeed, given such y, the allocation $x = y + \lambda\mathbf{1}$ satisfies $x(N) = T$, $x \geq \lambda\mathbf{1} \geq 0$ and $x(S) \leq f(S)$, hence lies in the core with floor at least λ. Note $f' \geq 0$ by definition of λ. By linear-programming duality,

$$\max\{y(N) :\ y \geq 0,\ y(S) \leq f'(S)\ \forall S\} = \min\Big\{\textstyle\sum_S w_S f'(S) :\ w \geq 0,\ \sum_{S\ni i} w_S \geq 1\ \forall i\Big\},$$

the minimum running over fractional covers. Taking $w = \mathbf{1}_{\{S=N\}}$ shows the minimum is at most $f'(N) = T - n\lambda$, so Conjecture 8 is exactly the assertion

$$\sum_S w_S f'(S)\ \geq\ f'(N) \qquad \text{for every fractional cover } w. \tag{2}$$

Proposition 9 (balanced covers)**.** *Inequality (2) holds for every exact (balanced) cover, i.e. whenever* $\sum_{S\ni i} w_S = 1$ *for all* i*.*

Proof. Balancedness gives $\sum_S w_S|S| = \sum_i \sum_{S\ni i} w_S = n$, so, using $f \geq g$ and the modularity of g,

$$\sum_S w_S f'(S) = \sum_S w_S f(S) - \lambda n\ \geq\ \sum_S w_S g(S) - \lambda n = \sum_i g_i \sum_{S\ni i} w_S - \lambda n = g(N) - \lambda n = f'(N),$$

the last step because $g(N) = T$ and $f'(N) = T - \lambda n$. The same computation works with h in place of g. □

What is missing is the case of covers that strictly over-cover some players. Writing $e_i := \sum_{S\ni i} w_S - 1 \geq 0$, the same computation gives

$$\sum_S w_S f'(S)\ \geq\ f'(N) + \max\Big(\sum_{i\in A} g_i e_i,\ \sum_{i\in B} h_i e_i\Big) - \lambda \sum_{i\in N} e_i,$$

and since $\lambda \leq f(\{i\})$, which equals g_i for $i \in A$ and h_i for $i \in B$, each inner sum dominates λ times its *own* block's excess but not the other block's. Closing that gap, or constructing the attaining core allocation directly, is the remaining work.

3.3 The leximax-min ceiling is not dual

The mirror-image statement is an inequality but not an identity, and this is the first place where the two selections part company.

Observation 10 (M-side asymmetry)**.** *For every core allocation, and in particular for* $\mathrm{Lmax}(v)$*,*

$$M\ \geq\ \max_{\substack{P\subseteq A,\ Q\subseteq B\\ (P,Q)\neq(\emptyset,\emptyset)}} \frac{\min\big(g(P), h(Q)\big)}{|P| + |Q|}. \tag{3}$$

Inequality (3) is not an identity: it fails strictly already at $n = 6$*.*

Proof of the inequality. Let $x \in \mathrm{Core}(v)$ and put $S = P \cup Q$. Then $g(S) = g(P)$ and $h(S) = h(Q)$ because g vanishes on B and h on A, so

$$\max_i x_i\ \geq\ \frac{x(S)}{|S|}\ \geq\ \frac{v(S)}{|S|} = \frac{\min\big(g(P), h(Q)\big)}{|P| + |Q|}.$$

Apply this to $x = \mathrm{Lmax}(v)$. Equivalently, (3) is Theorem 6 read on complements. □

How to read (3). The maximum runs over pairs (P, Q) other than $(\emptyset, \emptyset)$, with the convention $g(\emptyset) = h(\emptyset) = 0$. There is no infinity convention: a pair with an empty part contributes $\min(\cdot, 0) = 0$ and is simply inactive, so only pairs with *both* parts nonempty can bind. This is visible in the proof, where an empty part makes $v(S)$ the minimum of something and 0. Reading an empty part as $+\infty$ instead would reduce (3) to a maximum over singletons and give a different — and wrong — value; the misreading is easy enough that we spell the convention out.

The smallest counterexample. Scanning the complete two-generator enumeration in increasing T, no counterexample to equality in (3) exists at $n \le 5$ for any T, nor at $n = 6$ for $T \le 4$. At $n = 6$, $T = 5$:

$$g = (3, 2, 0, 0, 0, 0), \qquad h = (0, 0, 2, 1, 1, 1),$$
$$\mathrm{Lmin}(v) = \left(1, \tfrac{2}{3}, \tfrac{4}{3}, \tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right), \qquad \mathrm{Lmax}(v) = \left(\tfrac{6}{5}, \tfrac{4}{5}, \tfrac{6}{5}, \tfrac{3}{5}, \tfrac{3}{5}, \tfrac{3}{5}\right),$$

so $M = 6/5$, whereas the right-hand side of (3) equals 1, attained for instance at $P = \{1\}, Q = \{3\}$ with $\min(3, 2)/2 = 1$. The gap is $1/5$. One further counterexample class occurs at $n = 6$, $T = 5$: $g = (3, 1, 1)$, $h = (2, 2, 1)$ on the respective supports (its block-swapped relabelling is the same game up to permutation), again with $M = 6/5$ against a bound of 1. Over the same random samples used for Conjecture 8, (3) is tight in 120/120 games at $n = 4$, 218/218 at $n = 5$, 160/165 at $n = 6$ and 74/83 at $n = 7$.

Remark 11 (diagnosis)**.** *Theorem 7 bounds the floor using only the upper caps $x(S) \le f(S)$, and a single tightest cap can be saturated on its own, which is why one coalition certifies the optimum. Bounding the ceiling instead requires the lower bounds $x(N \setminus S) \ge T - f(S)$, and* minimizing *a maximum must respect several of them simultaneously; no single one certifies the optimum. This is a genuine asymmetry between the two selections, and it is the reason* Lmax *is the harder object throughout.*

4 The master family

Definition 12. *For integers $p \ge 1$ and $k \ge 2$ put $n = p + 1 + k$ and $T = pk$, and let $G(p, k)$ be the lower envelope of*

$$g = \big(p(k-1),\ \underbrace{1, \dots, 1}_{p},\ \underbrace{0, \dots, 0}_{k}\big), \qquad h = \big(\underbrace{0, \dots, 0}_{p+1},\ \underbrace{p, \dots, p}_{k}\big).$$

We index the players $0, 1, \dots, p + k$ and call player 0 the head*, $O := \{1, \dots, p\}$ the* ones*, and $B := \{p + 1, \dots, p + k\}$; so $A = \{0\} \cup O$ has $p + 1$ elements and $|B| = k$.*

Both generators are nonnegative with total $p(k-1)+p = pk = T$, their supports are disjoint, and every coordinate vanishes in one of them; by Lemma 1, $G(p, k)$ is exact and singleton-normalized.

Theorem 13 (closed forms)**.** *For all $p \ge 1$, $k \ge 2$,*

$$\mathrm{Lmin}\big(G(p,k)\big) = \left(\tfrac{p^2(k-1)}{p+1},\ \left[\tfrac{p}{p+1}\right]^{p+k}\right),$$
$$\mathrm{Lmax}\big(G(p,k)\big) = \left(\tfrac{p(k-1)}{k},\ \left[\tfrac{1}{k}\right]^{p},\ \left[\tfrac{p(k-1)}{k}\right]^{k}\right),$$
$$\big\|\mathrm{Lmin} - \mathrm{Lmax}\big\|_1 = \frac{2p\big(p(k-1)-1\big)}{p+1}, \qquad \rho\big(G(p,k)\big) = \frac{2\big(p(k-1)-1\big)}{k(p+1)},$$

where the exponents indicate the number of coordinates, listed in the order head, ones, B.

The proof occupies Sections 4.1–4.6: a key cap, the two optimizations, a membership check, and the arithmetic of the gap.

4.1 The key cap

For each $b \in B$,

$$f\big(O \cup \{b\}\big) = \max\big(g(O), h(\{b\})\big) = \max(p, p) = p,$$

so by Theorem 6 every core allocation x satisfies

$$x_1 + \cdots + x_p + x_b \;\le\; p \qquad \text{for every } b \in B. \tag{KC}$$

That single family of $(p+1)$-player caps drives everything below. Note that the two generators *tie* at these coalitions: the maximum defining f is attained on both sides.

4.2 Leximin

Let $x \in \mathrm{Core}(G(p,k))$ and $m := \min_i x_i$. Applying (KC) with any fixed b gives $(p+1)m \le p$, so

$$m \le \frac{p}{p+1}, \qquad \text{hence} \qquad \mu \le \frac{p}{p+1}.$$

Suppose $m = p/(p+1)$. Then in (KC) the $p+1$ numbers $x_1, \ldots, x_p, x_b$ are each at least $p/(p+1)$ and sum to at most $p = (p+1)\cdot p/(p+1)$, so all of them equal $p/(p+1)$. This holds for *every* $b \in B$, hence

$$x_1 = \cdots = x_p = x_{p+1} = \cdots = x_{p+k} = \frac{p}{p+1},$$

and efficiency forces the head to

$$x_0 = T - (p+k)\frac{p}{p+1} = pk - \frac{p(p+k)}{p+1} = \frac{p^2(k-1)}{p+1},$$

using $k - (p+k)/(p+1) = (kp+k-p-k)/(p+1) = p(k-1)/(p+1)$. Thus the max-min allocation is *unique*, and a unique max-min allocation is a fortiori the leximin allocation. Since $p(k-1) \ge 1$, the head's coordinate is at least $p/(p+1)$, so the floor of this allocation is indeed $p/(p+1)$. It remains to check that this point is in the core, which is done in Section 4.4.

4.3 Leximax-min

Let $x \in \mathrm{Core}(G(p,k))$, $M := \max_i x_i$ and $s := x_1 + \cdots + x_p$. By (KC), $x_b \le p - s$ for every $b \in B$, and trivially $x_b \le M$ and $x_0 \le M$. Efficiency gives

$$pk = x_0 + s + \sum_{b \in B} x_b \;\le\; M + s + k\min(M,\, p-s). \tag{4}$$

Case A: $M \le p - s$. Then (4) reads $pk \le (k+1)M + s$. The case hypothesis is $s \le p - M$, so $pk \le (k+1)M + p - M = kM + p$, i.e. $M \ge p(k-1)/k$.

Case B: $M > p - s$. Then (4) reads $pk \le M + s + k(p-s)$, i.e. $(k-1)s \le M$, so $s \le M/(k-1)$. The case hypothesis then gives $M > p - s \ge p - M/(k-1)$, i.e. $Mk/(k-1) > p$, i.e. $M > p(k-1)/k$.

In both cases $M \ge p(k-1)/k$, so the leximax-min ceiling is at least $p(k-1)/k$. Equality can occur only in Case A with every inequality used tight, that is with $x_0 = M$, with $x_b = M$ for all $b \in B$, and with

$$s = pk - (k+1)M = pk - (k+1)\frac{p(k-1)}{k} = \frac{p\big(k^2 - (k+1)(k-1)\big)}{k} = \frac{p}{k}.$$

(Consistency: $p - s = p - p/k = p(k-1)/k = M$, so Case A holds with equality, as it must.) Thus at the min-max level the head and all of B are pinned at $p(k-1)/k$ and the ones share exactly p/k; since $p/k \le p(k-1)/k$ for $k \ge 2$, the ones occupy the last p positions of the decreasing rearrangement. The next leximax stage minimizes their maximum subject to their sum being p/k, whose unique optimum is $1/k$ each. Hence Lmax is as claimed, again modulo membership.

4.4 Membership

This step is load-bearing: the two optimizations above determine the optima only conditionally on the described allocations being feasible.

Index as in Definition 12 and, for $S \subseteq N$, put

$$\varepsilon := [\,0 \in S\,], \qquad i := |S \cap O|, \qquad j := |S \cap B|,$$

so that $0 \le i \le p$, $0 \le j \le k$ and

$$g(S) = \varepsilon\, p(k-1) + i, \qquad h(S) = jp, \qquad f(S) = \max\big(\varepsilon\, p(k-1) + i,\ jp\big). \tag{5}$$

Lemma 14. *Both allocations of Theorem 13 lie in* $\mathrm{Core}(G(p,k))$ *for all* $p \ge 1$, $k \ge 2$. *Moreover, for both of them the tight coalitions are exactly*

$$\begin{array}{lll} \textit{(T1)} & S = O \cup \{b\},\ b \in B, & f(S) = \max(p,p) = p, \\ \textit{(T2)} & S = \{0\} \cup B',\ |B'| = k-1, & f(S) = \max\big(p(k-1),(k-1)p\big) = p(k-1), \\ \textit{(T3)} & S = N, & f(N) = T. \end{array}$$

Proof. The leximin allocation. Write $q := p/(p+1)$ and $H := p^2(k-1)/(p+1) = p(k-1)q$, so $x^* = (H, q, \dots, q)$ with $p+k$ copies of q. Efficiency:

$$H + (p+k)q = q\big(p(k-1) + p + k\big) = q(pk+k) = k(p+1)\frac{p}{p+1} = pk = T,$$

and $x^* > 0$. It remains to check $x^*(S) = \varepsilon H + (i+j)q \le f(S)$.

$\varepsilon = 0$, $j = 0$: S is a subset of O, and $x^*(S) = iq < i = f(S)$ for $i \ge 1$ because $q < 1$; for $i = 0$, $S = \emptyset$. Strict.

$\varepsilon = 0$, $j \ge 1$: bound $f(S) \ge jp$ and use $i \le p$:

$$x^*(S) = (i+j)\frac{p}{p+1} \le (p+j)\frac{p}{p+1} \le jp \iff p + j \le j(p+1) \iff p \le jp,$$

which holds since $j \ge 1$. Equality throughout requires $i = p$ and $j = 1$, i.e. $S = O \cup \{b\}$, which is (T1).

$\varepsilon = 1$, $j \le k-1$: put $G_0 := p(k-1) + i = g(S)$, so $x^*(S) = q(G_0 + j)$ and

$$x^*(S) \le G_0 \iff p(G_0 + j) \le (p+1)G_0 \iff jp \le G_0 = p(k-1) + i,$$

which holds because $jp \le (k-1)p$. Hence $x^*(S) \le g(S) \le f(S)$, with equality exactly when $j = k-1$ and $i = 0$, i.e. $S = \{0\} \cup B'$ with $|B'| = k-1$, which is (T2).

$\varepsilon = 1$, $j = k$: then $h(S) = kp = T$, so $f(S) \ge T \ge x^*(S)$, with equality only for $S = N$ since $x^* > 0$; this is (T3).

The leximax-min allocation. Write $L := p(k-1)/k$ and $r := 1/k$, so $y^* = (L, r, \dots, r, L, \dots, L)$ with p copies of r and k copies of L. Efficiency:

$$L + pr + kL = (k+1)L + \frac{p}{k} = \frac{p\big[(k+1)(k-1)+1\big]}{k} = \frac{pk^2}{k} = pk = T,$$

and $y^* > 0$. Now $y^*(S) = \varepsilon L + ir + jL$.

$\varepsilon = 0$, $j = 0$: $y^*(S) = i/k \le i = f(S)$ since $k \ge 2$, strictly for $i \ge 1$.

$\varepsilon = 0$, $j \ge 1$: comparing with the h-side,

$$\frac{i + jp(k-1)}{k} \le jp \iff i + jp(k-1) \le jpk \iff i \le jp,$$

which holds since $i \le p \le jp$, with equality exactly when $i = p$ and $j = 1$, i.e. (T1).

$\varepsilon = 1$, $j \le k-1$: $y^*(S) = (j+1)L + i/k$, and multiplying by $k > 0$,

$$y^*(S) \le p(k-1) + i \iff (j+1)p(k-1) + i \le kp(k-1) + ki \iff p(j+1-k) \le i,$$

after dividing by $k - 1 \ge 1$; the left-hand side is at most $0 \le i$, with equality exactly when $j = k-1$ and $i = 0$, i.e. (T2).

$\varepsilon = 1$, $j = k$: as before $f(S) \ge h(S) = kp = T \ge y^*(S)$, with equality only at $S = N$. □

4.5 The hinge: both selections saturate the same caps

Lemma 14 contains the expository point of the paper. The active constraints are the same for Lmin and Lmax: the two families (T1) and (T2), plus efficiency. Both are coalitions at which the two generators *tie*, which is the structural reason they are the active ones — there is no slack on either side to give. What differs is only how the two selections distribute mass *inside* a tight coalition. On $S = O \cup \{b\}$, which has total exactly p:

	each of the p ones	the single $b \in B$
Lmin :	$\frac{p}{p+1}$	$\frac{p}{p+1}$
Lmax :	$\frac{1}{k}$	$\frac{p(k-1)}{k}$

Leximin splits the budget p evenly over all $p+1$ members; leximax-min gives the ones the least it can and puts the whole remainder on the one member of B, because loading B is how it lowers the global maximum. The divergence is visible in this one coalition. Everything else is bookkeeping: leximin's even split leaves the residual $T - (p+k)p/(p+1)$ with nowhere to go but the head, while leximax-min's loading of every $b \in B$ up to $p(k-1)/k$ consumes almost all of T across B.

4.6 The gap

Proof of Theorem 13, conclusion. Put $D := p(k-1) - 1 \ge 0$. Coordinatewise,

$$\text{head:} \quad \frac{p^2(k-1)}{p+1} - \frac{p(k-1)}{k} = \frac{p(k-1)D}{k(p+1)},$$

$$\text{each of the } p \text{ ones:} \quad \frac{p}{p+1} - \frac{1}{k} = \frac{D}{k(p+1)},$$

$$\text{each of the } k \text{ members of } B\text{:} \quad \frac{p(k-1)}{k} - \frac{p}{p+1} = \frac{pD}{k(p+1)},$$

all three nonnegative. Hence

$$\| \text{Lmin} - \text{Lmax} \|_1 = \frac{p(k-1)D + pD + kpD}{k(p+1)} = \frac{D\big(p(k-1) + p + kp\big)}{k(p+1)} = \frac{2pkD}{k(p+1)} = \frac{2pD}{p+1},$$

and $\rho = 2pD/\big((p+1)pk\big) = 2D/\big(k(p+1)\big)$. □

Corollary 15 (deficit identity). $2-\rho\big(G(p,k)\big)=\dfrac{2n}{k(p+1)}$, *where* $n=p+1+k$.

Proof. $2-\dfrac{2\big(p(k-1)-1\big)}{k(p+1)}=\dfrac{2\big[k(p+1)-p(k-1)+1\big]}{k(p+1)}=\dfrac{2(kp+k-pk+p+1)}{k(p+1)}=\dfrac{2(k+p+1)}{k(p+1)}$. □

Remark 16 (the deficit is twice the overlap). *Corollary 15 also follows from Lemma 3. In* $G(p,k)$ *the coordinatewise minima of the two allocations are* $p(k-1)/k$ *at the head,* $1/k$ *at each one, and* $p/(p+1)$ *at each member of* B*, so*

$$\sum_i \min\big(\mathrm{Lmin}_i,\mathrm{Lmax}_i\big)=\frac{p(k-1)}{k}+\frac{p}{k}+\frac{kp}{p+1}=p+\frac{kp}{p+1}<p+k=n-1,$$

and $2\cdot\big(p+kp/(p+1)\big)/(pk)=2(p+1+k)/\big(k(p+1)\big)$. *The overlap between the two selections is thus bounded by* n *while* $T=pk$ *grows quadratically: this is the precise sense in which their supports become asymptotically disjoint.*

4.7 The supremum

Theorem 17 (value of the supremum). *For every* $n\ge 4$,

$$s(n)\;\ge\;2-\frac{8n}{n^2-(n \bmod 2)}=\begin{cases}2-\dfrac{8}{n}, & n \text{ even},\\[2mm] 2-\dfrac{8n}{n^2-1}, & n \text{ odd},\end{cases}$$

attained on the two-generator class by $G(p,k)$ *with the balanced split* $k=\lceil n/2\rceil$, $p+1=\lfloor n/2\rfloor$. *Consequently* $\sup_n s(n)=2$.

Proof. By Corollary 15 the deficit of $G(p,k)$ is $2n/(k(p+1))$ with $k+(p+1)=n$, $k\ge 2$ and $p+1\ge 2$. Minimizing the deficit means maximizing the product $k(p+1)$ of two positive integers with fixed sum n, whose maximum is $\lfloor n/2\rfloor\lceil n/2\rceil$, equal to $n^2/4$ for even n and $(n^2-1)/4$ for odd n, attained at the balanced split (for $n\ge 4$ this split satisfies the constraints $k\ge 2$, $p\ge 1$). Substituting gives the displayed bound. For the last claim, $\rho\le 2$ always by Lemma 3, and $2-8/n\to 2$. □

Table 1 lists the resulting values. The ratio reaches 1 at $n=8$ and passes it at $n=9$: from eight players on the two egalitarian core selections can differ by the entire worth of the grand coalition, and from nine players by more.

Remark 18 (the degenerate corner). $G(1,2)$ *has* $n=4$, $T=2$, $g=(1,1,0,0)$, $h=(0,0,1,1)$ *and* $\rho=2(1\cdot 1-1)/(2\cdot 2)=0$, *consistent with the four-player rigidity; the deficit identity reads* $2-0=2n/(k(p+1))=8/4=2$.

Remark 19 (there is nothing special about $|B|=4$). *An earlier version of this family fixed* $k=|B|=4$, *giving* $\rho=(3p-1)/\big(2(p+1)\big)\to 3/2$. *Corollary 15 shows that* k *is a free parameter and that the optimum is* $k\approx n/2$, *so the limit* $3/2$ *was an artifact of the fixed block size; the* $k=4$ *slice is already beaten from* $n=10$ *on (*$7/6$ *against* $6/5$*). The number* 4 *that appears as* $|\,\mathrm{supp}(\mathrm{Lmax})|$ *in the* $n=5$ *extremal class (Section 5) is a small-*n *phenomenon and not a feature of the asymptotics: in* $G(p,k)$ *every coordinate of* Lmax *is strictly positive, so* $|\,\mathrm{supp}(\mathrm{Lmax})|=n$ *and no player is starved at all.*

n	(p,k)	lower bound on $s(n)$	decimal
6	$(2,3)$	$2/3$	0.6667
7	$(2,4)$	$5/6$	0.8333
8	$(3,4)$	1	1.0000
9	$(3,5)$	$11/10$	1.1000
10	$(4,5)$	$6/5$	1.2000
11	$(4,6)$	$19/15$	1.2667
12	$(5,6)$	$4/3$	1.3333
16	$(7,8)$	$3/2$	1.5000
20	$(9,10)$	$8/5$	1.6000
100	$(49,50)$	$48/25$	1.9200

Table 1: The bound of Theorem 17 at the balanced split. All entries are exact; decimals are rounded.

n	totals	games	enumerated max ρ	Theorem 17
3	2–30	225	0	—
4	2–30	3,040	0	0
5	2–24	4,357	$1/2$	$1/3$
6	2–20	8,213	$2/3$	$2/3$
7	2–14	1,609	$5/6$	$5/6$
8	2–12	1,049	1	1

Table 2: Complete two-generator enumerations. The rows $n = 3, 4$ are controls: they reproduce the four-player rigidity. The last column is the value of the balanced-split family bound.

5 Fixed n: enumeration, certificates, and the $n = 5$ anomaly

5.1 Complete two-generator enumeration

Write $s_2(n)$ for the supremum of ρ over singleton-normalized two-generator exact games on n players. Table 2 reports complete enumerations, up to player permutation, generator swap and scaling, of all integer generator pairs with common total in the stated range. All arithmetic is exact rational.

Two readings must be kept apart.

What is complete. Because ρ is scale-invariant, every two-generator game with rational generators has a unique primitive integer representative, and enumeration to total $T_{\max}$ is exactly complete for the games whose primitive representative has total at most $T_{\max}$ — equivalently, for rational shapes of denominator at most $T_{\max}$. It says nothing about finer denominators or about irrational generators, so enumeration-to-T is a lower-bound method for $s_2(n)$: the entries of Table 2 are maxima over the enumerated range, not values of s_2. As a partial probe of the excluded region, shape hill-climbing at fine resolution was run from the known witnesses (resolution $1/60$ at $n = 5$; $1/56$ and $1/60$ at $n = 6$; $1/64$ at $n = 7$). None improved on the tabulated maximum; at $n = 7$ the climb from the $5/6$ witness stalled after 42 steps, and a second start from a base of $2/3$ climbed only to $3/4$.

What agrees. The enumerated maxima agree exactly with Theorem 17 at $n = 6, 7, 8$, so on the two-generator class that bound is attained there, and

$$s_2(n) \stackrel{?}{=} 2 - \frac{8n}{n^2 - (n \bmod 2)} \qquad (n \geq 6)$$

is a live candidate, currently confirmed at three values of n. At $n = 5$ the two disagree, and the enumeration is the correct one; see Section 5.3.

5.2 Searching the full exact class

The two-generator restriction is not innocuous. Adding a generator *lowers* the envelope and hence *grows* the core, so it is not automatic that two generators are extremal. The following observation lets one search the entire exact class rather than a bounded-generator slice.

Lemma 20 (core closing)**.** *Fix $T > 0$ and a vector $c = (c_S)$ indexed by the proper coalitions, and put $P(c) = \{x : x(N) = T,\ x(S) \geq c_S$ for all proper $S\}$. If $P(c) \neq \emptyset$, then the game v defined by $v(S) = \min_{x \in P(c)} x(S)$ for proper S and $v(N) = T$ is exact, and $\mathrm{Core}(v) = P(c)$.*

Proof. Every $x \in P(c)$ satisfies $x(S) \geq v(S)$ by definition of v and $x(N) = T$, so $P(c) \subseteq \mathrm{Core}(v)$; and each minimum is attained, so v is exact. Conversely, $v(S) \geq c_S$ for every proper S because every point of $P(c)$ satisfies $x(S) \geq c_S$; hence $x \in \mathrm{Core}(v)$ implies $x(S) \geq v(S) \geq c_S$ and $x(N) = T$, i.e. $x \in P(c)$. □

Perturbing c and re-closing therefore moves within the exact class and reaches cores with arbitrarily many vertices. Hill climbing in this "c-space", *seeded at the extremal two-generator game* of Table 2, produced no improvement at $n = 5$ (240 steps, stays at $1/2$), $n = 6$ (496 steps, stays at $2/3$) or $n = 7$ (301 steps, stays at $5/6$). Since the search is over all exact games, this certifies the two-generator optimum as a *local* optimum in the full exact class at those three values of n. Independently, from the known extremal at each of $n = 5, 6, 7, 8$, about 200 structured and random third generators at fixed common total were added: zero improvements.

One limitation must be stated. Unseeded c-space restarts reach only $1/3$ at $n = 6$, far below the known $2/3$; that variant of the search is under-powered for $n \geq 6$ and certifies nothing. Only the seeded runs support the local-optimality claim, and none of these runs bears on global optimality.

5.3 The $n = 5$ anomaly

At $n = 5$ the balanced-split family gives only $1/3$, but the truth is $1/2$, and the witness has a non-uniform h-block. The proof is by hand from two flat splits of the game; no linear program and no vertex enumeration is used.

Theorem 21 ($n = 5$)**.** *Let w be the lower envelope of $g^1 = (1, 0, 0, 1, 2)$ and $g^2 = (0, 1, 3, 0, 0)$. Then w is exact and singleton-normalized with $w(N) = 4$, and*

$$\mathrm{Lmin}(w) = \left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{5}{4}, \tfrac{1}{2}, \tfrac{5}{4}\right), \qquad \mathrm{Lmax}(w) = (1, 0, 1, 1, 1), \qquad \| \mathrm{Lmin} - \mathrm{Lmax} \|_1 = 2, \qquad \rho(w) = \tfrac{1}{2}.$$

In particular $s(5) \geq 1/2 > 1/3$.

Proof. Exactness and normalization are immediate from Lemma 1, each coordinate vanishing in one generator. Two flat splits control the core: $w(\{1, 2\}) = 1$ and $w(\{3, 4, 5\}) = 3$ sum to $w(N) = 4$, so

$x_1 + x_2 = 1$ on all of Core(w), and likewise $w(\{2,4\}) = 1$ with $w(\{1,3,5\}) = 3$ gives $x_2 + x_4 = 1$. Putting $s := x_1$, every core point therefore satisfies

$$x_1 = x_4 = s, \qquad x_2 = 1 - s, \qquad x_3 + x_5 = 3 - s, \tag{6}$$

with $s \in [0,1]$ by nonnegativity of x_1 and x_2.

Leximin. By (6), $\min_i x_i \le \min(s, 1-s) \le 1/2$, with equality forcing $s = 1/2$ and hence $x_1 = x_2 = x_4 = 1/2$. Then $x_3 + x_5 = 5/2$, and the leximin continuation splits this evenly, giving $5/4$ each. The resulting allocation $x = (1/2, 1/2, 5/4, 1/2, 5/4)$ lies in the core, as direct substitution against both generators confirms. Hence Lmin(w) is as claimed.

Leximax-min. By (6), $\max_i x_i \ge (x_3 + x_5)/2 = (3-s)/2 \ge 1$ since $s \le 1$. Equality forces $s = 1$, so $x = (1, 0, x_3, 1, x_5)$ with $x_3 + x_5 = 2$; the ceiling $\max_i x_i = 1$ then forces $x_3 = x_5 = 1$. The unique candidate $(1,0,1,1,1)$ lies in the core, so Lmax(w) $= (1,0,1,1,1)$.

Finally $\|\,\text{Lmin} - \text{Lmax}\,\|_1 = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 2$ and $\rho = 2/4 = 1/2$. □

Four independent methods agree on this game: the hand argument above, exact rational core-vertex enumeration, a staged exact solver over linear real arithmetic, and a from-scratch exact-rational two-phase simplex with Bland's rule.

Conjecture 22 ($n = 5$ supremum)**.** *$s(5) = 1/2$.*

Evidence. Roughly 4×10^4 distinct exact five-player games were evaluated in exact rational arithmetic with nothing above $1/2$: the complete two-generator enumeration of Table 2 (4,357 games, maximum $1/2$); complete three-generator enumerations at totals $4, 5, 6, 7$ (335, 1,175, 3,747 and 9,923 games, maxima $1/2$, $2/5$, $1/3$, $8/21$); c-space hill climbing in the sense of Lemma 20 (68 restarts, about 6,600 evaluations), which plateaus at $1/2$; and fine-denominator shape climbing from the extremal witness scaled to total 60, which does not improve. The controls at $n = 3, 4$ in Table 2 return maximum 0 from the same pipeline.

Structure of the extremal class. The following are computational findings over 360 distinct profiled games at $n = 5$, of which 199 have $\rho > 1/3$ and 107 attain $\rho = 1/2$. Among the 107 extremal games, without exception:

- Lmin has stratum composition $(3,2)$, and Lmax has stratum composition $(1,4)$;
- Lmax has exactly one zero coordinate, so $|\,\text{supp}(\text{Lmax})| = 4$;
- normalized to $T = 4$ the pair is unique up to permutation: Lmin $\sim (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{5}{4}, \frac{5}{4})$ and Lmax $\sim (0,1,1,1,1)$ — that is, exactly the pair of Theorem 21.

Among the 199 games above $1/3$, the Lmin strata are $(3,2)$ in 196 cases and $(2,1,2)$ in 3; the pattern $(4,1)$ never occurs above $1/3$, and Lmax is never the uniform allocation.

One tempting invariant is refuted. Call S a *flat split* if $v(S) + v(N \setminus S) = v(N)$; then $x(S)$ is constant on the core, which is the affine seesaw visible in (6) — in the game of Theorem 21, $w(\{1,2\}) + w(\{3,4,5\}) = 1 + 3 = 4$. Only 76 of the 107 extremal games have any flat split, and 31 attain exactly $1/2$ with none, so the seesaw is a feature of the particular presentation in Theorem 21 and not an invariant of extremality. When a flat split does exist its sizes are always $\{2,3\}$, and the Lmax-starved player lies on the size-2 side in all 76 cases.

Two regimes. Theorem 17 gives only $1/3$ at $n = 5$ because the family $G(p,k)$ has a uniform h-block and the available splits $(k, p+1) \in \{(2,3),(3,2)\}$ are too coarse; the extremal five-player game has a non-uniform h-block. The crossover to the G-family shape happens at $n = 6$. The two regimes differ qualitatively and not just numerically: at $n = 5$ the extremal Lmax starves one

player to 0 and has support of size 4, whereas in $G(p,k)$ the extremal Lmax is strictly positive everywhere. Explaining the crossover is open (Section 7).

5.4 One-coalition certificates and the regime change

Write, for any singleton-normalized exact game with the general dual bound $f(S) = T - v(N \setminus S)$,

$$\lambda := \min_{\emptyset \neq S} \frac{f(S)}{|S|}, \qquad \nu := \max_{\emptyset \neq S} \frac{v(S)}{|S|}, \qquad \mu := \min_i \mathrm{Lmin}(v)_i, \qquad M := \max_i \mathrm{Lmax}(v)_i.$$

The general duality gives the bracketing $\mu \leq \lambda$ and $M \geq \nu$ at once (for $\mu \leq \lambda$ the proof of Theorem 7 applies verbatim). On a two-generator game the right-hand side of (3) is exactly ν, since $v(P \cup Q) = \min(g(P), h(Q))$. Observation 10 says the second bracket can be strict from $n = 6$ on. Two facts sharpen this into a statement about the regime change at five players.

Proposition 23 (covering obstruction)**.** *Let $\tau_\nu := \{S : v(S) = \nu|S|\}$ be the ν-tight family. If τ_ν covers N and $\nu > T/n$, then no core point satisfies $x \leq \nu\mathbf{1}$, and hence $M > \nu$.*

Proof. Sum the core inequalities $x(S) \geq v(S) = \nu|S|$ over a cover with weights $w \geq 0$: with $c_i := \sum_{S \ni i} w_S \geq 1$ this gives $\sum_i c_i(x_i - \nu) \geq 0$. If $x \leq \nu\mathbf{1}$, every term is nonpositive, so $x_i = \nu$ for every i; then $T = \nu n$, contradicting $\nu > T/n$. □

Moreover no *balanced* collection can witness $M > \nu$: for balanced w, $\sum_S w_S v(S) \leq T$ because the core is nonempty, and the same computation then yields $\sum_S w_S\big(\nu|S| - v(S)\big) \geq \nu n - T$, which is what equality requires. So a failure of $M = \nu$ needs a strictly over-covering fractional cover: no balanced collection can witness one. A covering ν-tight family is one way this arises, and it is what occurs at the smallest counterexample of Observation 10: the four ν-tight sets $\{2,3\}, \{1,3,4\}, \{1,3,5\}, \{1,3,6\}$ cover N with per-player excess $(2,0,3,0,0,0)$, and player 2, lying only in $\{2,3\}$, blocks every balanced subfamily. At the five-player extremal game of Theorem 21, by contrast, $\nu = 1$ and the ν-tight family $\{3,5\}, \{1,3,5\}, \{3,4,5\}$ misses player 2, so Proposition 23 does not apply, and the dual minimum $h(N) = 1$ is attained at the balanced partition $\{2,4\} \mid \{1,3,5\}$. The covering test is sufficient for $M > \nu$ but not necessary: at six players, games with $M > \nu$ outnumber those with a covering ν-tight family.

Theorem 24 (one-coalition certificates at five players)**.** *For every singleton-normalized exact game on five players, $\mu = \lambda$ and $M = \nu$: both egalitarian levels are certified by a single coalition.*

Proof, by finite exact check. By the duality above, $M = \nu$ fails if and only if some fractional cover w has $\sum_S w_S h(S) < h(N)$, where $h(S) = \nu|S| - v(S) \geq 0$; likewise for μ with $f'(S) = f(S) - \lambda|S|$. Both objectives are nonnegative on the cover polyhedron, which is pointed with recession cone the nonnegative orthant, so each minimum is attained at a vertex; on five players there are 2287 vertices, each of support at most five. Fixing a vertex and a coalition attaining ν (resp. λ) makes every hypothesis, and the failure inequality, linear in v, and $v(N) = 1$ is free by homogeneity. Exactness is imposed through the 280 inequalities that the facet theorem of Studený and Kratochvíl (2022) assigns to the sixteen indecomposable min-semi-balanced systems on five players under all permutations; this description was validated in both directions against independent exactness tests. Each of the 1142 symmetry classes of (vertex, anchor) pairs, on each side, is infeasible in exact rational arithmetic. □

Two remarks calibrate the proof. First, deleting the 280 exact-cone inequalities makes 361 (resp. 339) of the over-covering classes feasible, so it is exactness — not the shape of the certificate

— that forces the one-coalition property at five players. Second, the M-side is false from six players on (Observation 10), and the same reduction detects that failure. The theorem is therefore the formal content of the regime change of Section 5.3: at five players both selections are governed by single bottleneck coalitions, while from six players the ceiling requires genuinely fractional covers — which is precisely where $s(n)$ begins its climb toward 2. One consequence localizes all the difficulty of $s(5)$.

Corollary 25 (floor from ceiling)**.** *For every singleton-normalized exact five-player game,* $\mu \geq \min_i \mathrm{Lmax}(v)_i$.

Proof. Lmax is a core point, so $f(S) = T - v(N \setminus S) \geq T - \mathrm{Lmax}(N \setminus S) = \mathrm{Lmax}(S)$ for every S; hence $\lambda \geq \min_S \mathrm{Lmax}(S)/|S| \geq \min_i \mathrm{Lmax}_i$, and $\mu = \lambda$ by Theorem 24. □

If Lmax has no zero coordinate, the leximin floor is therefore bounded below for free and elementary gap arithmetic bounds the ratio; every genuinely hard configuration for $s(5)$ has a *zero* in Lmax. The theorem does *not* by itself deliver the starvation bound needed for those configurations; that bound is a statement about a different functional and requires its own argument, which the next theorem supplies in the extremal one.

Theorem 26 (the extremal configuration is pinned)**.** *Let v be a singleton-normalized exact five-player game whose* Lmin *strata form the pattern* $(3, 2)$ *and whose* Lmax *strata form the pattern* $(1, 4)$ *with the singleton stratum at zero, at player j. Then*

$$f(\{j\}) = T - v(N \setminus \{j\}) \;\geq\; \tfrac{T}{8}.$$

Proof, by finite exact check. Suppose $f(\{j\}) < T/8$. By the size bound from the Lmax vector's core membership, every other coalition has $f(S)/|S| \geq T/8$, so Theorem 24 places the leximin floor at $\mu = f(\{j\}) < T/8$; and j is poor, since were j rich, then $f(\{j\}) \geq \mathrm{Lmin}_j = (T - 3\mu)/2 = (T - 3f(\{j\}))/2$ would give $f(\{j\}) \geq T/5 > T/8$, contradicting the failure hypothesis. The $(3, 2)$ hypothesis says the entire poor stratum is pinned at μ; by LP duality that pinning is certified by one of 354 vertices of a dual polyhedron involving no game data. Fixing the certificate and an anchor coalition makes everything linear in (v, μ), and all 10,974 resulting exact programs are infeasible. An ablation shows the check already goes through from a strictly smaller hypothesis set — core membership of the Lmax vector, core membership of the Lmin vector, and the pinning — so a direct argument is available that uses neither the facet description nor the M-side of Theorem 24. □

Corollary 27 (the pinned solution pair)**.** *In the configuration of Theorem 26, suppose in addition that the starved player is* Lmin*-poor and that some coalition attaining ν has three players. Then, up to relabelling,* $\mathrm{Lmin} = \left(\frac{T}{8}, \frac{T}{8}, \frac{T}{8}, \frac{5T}{16}, \frac{5T}{16}\right)$, $\mathrm{Lmax} = \left(0, \frac{T}{4}, \frac{T}{4}, \frac{T}{4}, \frac{T}{4}\right)$, *and* $\rho(v) = \frac{1}{2}$ *exactly.*

Proof. Normalize $T = 1$. The Lmax pattern gives $M = 1/4$ directly. Core membership of the Lmax vector gives, for $S \neq \{j\}$, $f(S) \geq \big(|S| - [\,j \in S\,]\big)/4 \geq |S|/8$, and Theorem 26 gives $f(\{j\}) \geq 1/8$, so $\lambda \geq 1/8$. By the M-side of Theorem 24, $\nu = M = 1/4$, so a three-player coalition S_0 attaining ν has $v(S_0) = 3/4$ and $f(N \setminus S_0) = 1/4$ on two players, so $\lambda \leq 1/8$; hence $\lambda = 1/8$, and Theorem 24 gives $\mu = 1/8$. With the starved player Lmin-poor, $\mu = 1/8 \leq 1/6$ gives $(1 - 3\mu)/2 \geq 1/4 = M$, so the coordinatewise gap telescopes to $\|\,\mathrm{Lmin} - \mathrm{Lmax}\,\|_1 = T - 4\mu = T/2$, and the two displayed allocations are the unique ones consistent with the strata, the floor and the totals. □

Both additional hypotheses hold in every configuration game in the combined harvests (53 of 53), and a wider profiling of games with the $(1, 4)$ Lmax pattern never produced an Lmin-rich starved player (0 of 68). Inside the proof of Theorem 26 the Lmin-poor property is derived from

the failure assumption, which is all that theorem needs; whether either hypothesis can fail in the configuration at all is open and recorded in Section 7.

For the ratio *bound*, as opposed to the pinned pair, both hypotheses can in fact be dispensed with.

Theorem 28 (unconditional configuration bound)**.** *In the configuration of Theorem 26,* $\rho(v) \le \frac{1}{2}$.

Proof, by gap arithmetic and one further finite check. Normalize $T = 1$; the Lmax pattern gives $M = 1/4$ and the $(3,2)$ pattern gives $\mathrm{Lmin} = (\mu, \mu, \mu, \varrho, \varrho)$ with $\varrho = (1 - 3\mu)/2$. If the starved player is Lmin-poor, the coordinatewise gap is $1-4\mu$ for $\mu \le 1/6$ and 2μ otherwise, and Theorem 26 gives $\mu \ge 1/8$, so both branches are at most $\frac{1}{2}$. If the starved player is Lmin-rich, the gap is $3/2-6\mu$ for $\mu \le 1/6$ and $1 - 3\mu$ otherwise, so the bound holds exactly when $\mu \ge 1/6$ — and a finite check of the same design as Theorem 26 (the rich relabelling of the pinning polyhedron; 31×354 exact programs, all infeasible, with the drop-the-pinning control restoring feasibility) shows $\mu < 1/6$ is impossible in that branch. □

Two calibrations. First, the finite check is not removable in favour of soft arguments: from Theorem 24 and core membership of both allocation vectors alone — everything that is game-free — the best obtainable bound on ρ at five players is exactly $\frac{3}{2}$, so every configuration needs a lower bound on the floor of the strength of Theorem 26. Second, the attempted refutation of the ν-triple hypothesis produced, once again, a certificate whose audited model is a genuine exact game *outside* the configuration (a $(1,4)$-shaped Lmax with minimum $1/30$, ratio $5/12$) — the third certificate-not-realizable instance of this program, and the reason hypotheses are retired here only by finite checks, never by feasible certificates alone.

The configuration hypothesis is essential, not decorative: dropping the $(3,2)$ pinning makes the same programs satisfiable, and their models are genuine exact five-player games with $T - v(N\setminus\{i\}) = T/16$ for some player i — half the bound — but with Lmin = Lmax, hence no separation at all. What remains open for $s(5) = \frac{1}{2}$ in full are the cases outside Corollary 27: the Lmax stratum shapes $(1,2,2)$ and $(1,1,3)$; the $(4,1)$ Lmin boundary; and whether the configuration can occur with an Lmin-rich starved player (never observed, 0 of 68). In the enumerations the first two never attain $\frac{1}{2}$ and the boundary case never exceeds $\frac{1}{3}$.

6 Obstructions to an upper bound

6.1 No constant below 2

Theorem 17 already rules out one shape of answer. Any bound of the form $\|\mathrm{Lmin}(v) - \mathrm{Lmax}(v)\|_1 \le c \cdot v(N)$ with a constant $c < 2$, valid for all n, is false: choose n even with $2 - 8/n > c$. So the only possible clean upper bound is dimension-dependent, and the right target is the finite-n statement

$$s_2(n) = 2 - \frac{8n}{n^2 - (n \bmod 2)} \qquad (n \ge 6),$$

currently confirmed by complete enumeration only at $n = 6, 7, 8$, together with the question of whether $s(n) = s_2(n)$.

6.2 No bound linear in the leximin floor

A second shape is also excluded, and this one was a live candidate. The inequality

$$\|\mathrm{Lmin}(v) - \mathrm{Lmax}(v)\|_1 \;\le\; 4(n-4)\,\mu(v), \qquad \mu(v) = \min_i \mathrm{Lmin}(v)_i,$$

holds with no violations in 900 fresh random five-player games and is tight both on the $n = 5$ extremal class and on $G(3, 4)$ at $n = 8$, where both sides equal 12. It fails from $n = 9$ on:

game	n	$\| \mathrm{Lmin} - \mathrm{Lmax} \|_1$	$4(n-4)\mu$
$G(4,4)$	9	88/5	16
$G(3,5)$	9	33/2	15
$G(4,5)$	10	24	96/5
$G(5,5)$	11	95/3	70/3

The reason is structural rather than accidental: in $G(p, k)$ one has $\mu = p/(p + 1) < 1$, bounded, while the gap grows like $2T$. No bound of the form $C(n)\mu(v)$ can track the divergence, whatever C is.

6.3 Non-monotonicity under majorization

The most natural route to an upper bound on the two-generator class is to argue that a uniform h-block is without loss of generality — the extremal family $G(p, k)$ has one — and then to optimize over g alone using the floor formula of Conjecture 8. That route is closed.

Define a *smoothing step* as a Robin Hood transfer of one unit from a larger to a smaller h-coordinate, so that h moves down in the majorization order, toward uniform. The premise that a smoothing step never decreases ρ is false, and not only for small blocks:

$\lvert B \rvert$	2	3	4	5
smoothing steps tested	40	76	84	84
violations	18	27	11	21
rate	45%	36%	13%	25%

In particular it fails for $|B| \geq 4$, so the block-size threshold one might guess from the $n = 5$ anomaly does not exist. An explicit verified counterexample at $|B| = 4$, with $n = 7$ and $T = 10$, both allocations confirmed in the core:

$$g = (0, 5, 0, 3, 2, 0, 0), \qquad h = (1, 0, 2, 0, 0, 5, 2),$$

$$\mathrm{Lmin} = (1, 2, 1, 1, 1, 3, 1), \qquad \mathrm{Lmax} = \left(\tfrac{5}{6}, \tfrac{5}{2}, \tfrac{5}{6}, \tfrac{4}{3}, \tfrac{7}{6}, \tfrac{5}{2}, \tfrac{5}{6}\right), \qquad \rho = \tfrac{1}{5}.$$

Smoothing the h-block $(5, 2) \mapsto (4, 3)$, i.e. replacing h by $(1, 0, 2, 0, 0, 4, 3)$, gives

$$\mathrm{Lmin} = \mathrm{Lmax} = (1, 2, 1, 1, 1, 2, 2), \qquad \rho = 0.$$

So a single Robin Hood step inside one generator collapses the ratio from 1/5 to 0, and ρ is not monotone under majorization of h.

The two facts sit together uncomfortably and both are true: smoothing makes a game more symmetric and tends to *collapse* the two selections onto each other, yet the extremal family has a perfectly uniform h. The dependence is therefore genuinely non-monotone, and any upper-bound proof must handle general h directly rather than reducing to the uniform case.

7 What the results do and do not establish

Proved. Lemma 1 (lower-envelope generation and its converse); Lemma 3 (ceiling and overlap identity); Lemma 4 (support disjointness); Theorem 6 (two-generator core duality, with identity (1) valid without disjointness); Theorem 7 ($\mu \le \lambda$); Proposition 9 (the cover inequality (2) for balanced covers); the validity half of Observation 10; Theorem 13 with Lemma 14 (closed forms for $G(p,k)$, including membership for all $p \ge 1$, $k \ge 2$ and the exact list of tight coalitions); Corollary 15 and Remark 16 (the deficit identity, twice); Theorem 17 ($s(n) \ge 2 - 8n/(n^2 - (n \bmod 2))$ and $\sup_n s(n) = 2$); Lemma 20 (core closing); Proposition 23 (the covering obstruction); Theorem 21 ($\rho = 1/2$ at $n = 5$, by hand). The failure of the $4(n-4)\mu$ bound from $n = 9$, and the non-monotonicity of ρ under majorization of h, are proved by the explicit games displayed in Section 6: each is a finite exact-rational computation on a named game, and each is a refutation, so a single instance suffices.

Proved by exhaustive finite check. Theorem 24 ($\mu = \lambda$ and $M = \nu$ for every exact five-player game; 1142 symmetry classes, each infeasible in exact rational arithmetic) and Theorem 26 with Corollary 27 ($f(\{j\}) \ge T/8$ in the extremal configuration; 10,974 programs); Theorem 28's rich-branch floor bound ($\mu \ge T/6$; another 10,974 programs). The checks are proof dependencies; the corollary's two additional hypotheses are open in general and listed under (4) below.

Enumerated, not proved. The entries of Table 2 are maxima over the stated integer ranges, hence complete only for rational shapes whose primitive integer representative has total in that range; they are lower bounds for $s_2(n)$, not its value. The strict failure of equality in (3) is established by the displayed six-player game, but the claim that this is the *smallest* such failure rests on the enumeration. The tightness rates for (3) and for Conjecture 8 are sample statistics over 586 random games. The structural findings about the $n = 5$ extremal class in Section 5.3 — unique solution pair up to permutation, $(3,2)$ and $(1,4)$ stratum compositions, $|\operatorname{supp}(\mathrm{Lmax})| = 4$, the refutation of the seesaw hypothesis — are computational, over 360 profiled games, and none of them is proved. The local-optimality certificates from seeded c-space climbing at $n = 5, 6, 7$ are numerical searches over the full exact class; the unseeded variant is under-powered at $n \ge 6$ and certifies nothing.

Conjectured. Five problems remain open.

(1) **Conjecture 8**, equality in the leximin floor formula. Reduced by linear-programming duality to the fractional-cover inequality (2); proved for balanced covers (Proposition 9); the over-covering case is open in general. At five players Theorem 24 *proves* the equality, for every exact game and not only two-generator ones. Verified in 586/586 random two-generator games at $n = 4, \ldots, 7$.

(2) **An upper bound.** Theorem 17 forces any true bound to be asymptotically 2, so every bound $c \cdot v(N)$ with $c < 2$ is false for large n. The right target is $s_2(n) = 2 - 8n/(n^2 - (n \bmod 2))$ for $n \ge 6$, confirmed by complete enumeration at $n = 6, 7, 8$ only. Section 6 closes the majorization route and the $C(n)\mu$ route.

(3) **Is the two-generator class extremal?** Adding a generator lowers the envelope and grows the core, so this is not automatic. All evidence is negative for a counterexample: zero improvements from about 200 third generators at each of $n = 5, 6, 7, 8$, and no improvement from seeded c-space climbing over the full exact class at $n = 5, 6, 7$. Open globally.

(4) **Conjecture 22**, $s(5) = 1/2$, together with the two-regime picture suggested by the extremal-class structure: $(3,2)$ Lmin-strata above $1/3$, and $(4,1)$ never exceeding $1/3$. The extremal

configuration is closed unconditionally for the bound (Theorem 28); its two hypotheses survive only in the sharper pinned-pair statement of Corollary 27. By Corollary 25 the remaining hard cases are exactly the configurations whose Lmax has a zero coordinate — the $(1, 2, 2)$, $(1, 1, 3)$ and $(2, 1, 2)$ shapes with vanishing bottom stratum — together with the $(4, 1)$ Lmin regime, whose analysis needs ceiling-side second-stage certificates that have been sketched but not built (the derivation is archived with the verifiers).

(5) **A structural explanation of the $n = 5$ versus $n \geq 6$ regime change**: a starved Lmax with $|\mathrm{supp}(\mathrm{Lmax})| = 4$ at $n = 5$ against a strictly positive Lmax in $G(p, k)$. Theorem 24 gives the change its formal shape — one-coalition certificates live and die at the threshold — but the question of *why five* remains open.

Not claimed. Exactness theory, lower-envelope exactification, and the characterizations of the egalitarian selections are prior art (Arin and Iñarra, 2001; Arin et al., 2003; Biswas et al., 1999; Dietzenbacher, 2021; Llerena and Mauri, 2017; Csóka et al., 2011; Studený and Kratochvíl, 2022); the four-player rigidity is the companion paper's classification (Flanagan, 2026) combined with the Lorenz implication of Llerena and Mauri (2017). The candidate new content is the two-generator core duality and its one-sided floor certificate, the family $G(p, k)$ with its closed forms and deficit identity, the resulting evaluation $\sup_n s(n) = 2$, the identification of the $n = 5$ anomaly with a proof by hand, the one-coalition certificate theorem and the pinned extremal configuration at five players, and the two closed routes of Section 6. Novelty remains conditional on specialist review.

Reproducibility note

Sections 2–4 and 6 are proved by hand and depend on no program. Theorems 24 and 26 are proved by exhaustive finite exact-rational checks, described in their proofs; those checks are proof dependencies, archived in `../verifiers` with the theorem-to-driver map in its README. The remaining programs are bug audits, not proof dependencies. Exact rational programs in the companion repository independently reconstruct all displayed games and coalition values, the closed forms of Theorem 13 at twelve parameter pairs, the full coalition-inequality check and the tight-set prediction of Lemma 14 for every (p, k) with $1 \leq p \leq 7$, $2 \leq k \leq 7$ and $n \leq 13$, and the four-way agreement on the game of Theorem 21. They are bug audits, not proof dependencies. All arithmetic is exact rational; no floating-point value appears in any stated result. Reported optima come from a from-scratch exact-rational two-phase simplex with Bland's rule, from a staged exact solver over linear real arithmetic, or from exact vertex enumeration, with at least two methods agreeing on every displayed allocation; one library simplex implementation cycles on these degenerate programs and was not used for any reported number.

References

Arin J, Iñarra E (2001) Egalitarian solutions in the core. International Journal of Game Theory 30:187–193.

Arin J, Kuipers J, Vermeulen D (2003) Some characterizations of egalitarian solutions on classes of TU-games. Mathematical Social Sciences 46:327–345. https://doi.org/10.1016/S0165-4896(03)00048-6

Biswas AK, Parthasarathy T, Potters JAM, Voorneveld M (1999) Large cores and exactness. Games and Economic Behavior 28:1–12. https://doi.org/10.1006/game.1998.0683

Csóka P, Herings PJJ, Kóczy LÁ (2011) Balancedness conditions for exact games. Mathematical Methods of Operations Research 74:41–52.

Fujishige S (1980) Lexicographically optimal base of a polymatroid with respect to a weight vector. Mathematics of Operations Research 5:186–196.

Gangam RR, Garg N, Shahkar P, Vazirani VV (2024) Equitable core imputations for max-flow, MST and b-matching games. Preprint, arXiv:2403.06037.

Klijn F, Vermeulen D, Hamers H, Solymosi T, Tijs S, Villar JP (2003) Neighbor games and the leximax solution. Mathematical Methods of Operations Research 58:191–208.

Llerena F, Mauri L (2016) Reduced games and egalitarian solutions. International Journal of Game Theory 45:1053–1069.

Vazirani VV (2024) Fair core imputations for the assignment game: new solution concepts and efficient algorithms. Preprint, arXiv:2402.11437.

Dutta B, Ray D (1989) A concept of egalitarianism under participation constraints. Econometrica 57:615–635.

Dietzenbacher B (2021) Monotonicity and egalitarianism. Games and Economic Behavior 127:194–205. https://doi.org/10.1016/j.geb.2021.03.006

Flanagan M (2026) Balanced transport at the egalitarian core: a complete boundary phase diagram. Preprint; companion paper.

Llerena F, Mauri L (2017) On the existence of the Dutta–Ray's egalitarian solution. Mathematical Social Sciences 89:92–99. https://doi.org/10.1016/j.mathsocsci.2017.07.002

Studený M, Kratochvíl V (2022) Facets of the cone of exact games. Mathematical Methods of Operations Research 95:35–80. https://doi.org/10.1007/s00186-022-00770-4